



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (E&E) (CBCS) Degree Examinations

December 2024

EE3242-1- VLSI CIRCUITS AND DESIGN

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **CMOS technology is used in developing which of the following?**
A) Microprocessors
B) Microcontrollers
C) DSP processors
D) All mentioned
2. **P-well is created on _____**
A) p substrate
B) n substrate
C) Both p and n substrate
D) None mentioned
3. **Lithography is a process of -----**
A) used to transfer a pattern to a layer on the chip
B) used to develop an oxidation layer on the chip
C) used to develop a metal layer on the chip
D) used to produce the chip
4. **In CMOS fabrication, the photoresist layer is exposed to _____**
A) visible light
B) ultraviolet light
C) infra red light
D) Fluorescent
5. **I_{ds} depends on**
A) V_g
B) V_{ds}
C) V_{dd}
D) V_{ss}
6. **According to body effect, substrate is biased with respect to**
A) source
B) Drain
C) gate
D) V_{ss}
7. **Transconductance gives the relationship between**
A) input current and output voltage
B) output current and input voltage
C) input current and input voltage
D) output current and output voltage
8. **The work function difference is negative for**
A) silicon substrate
B) polysilicon gate
C) both of the mentioned
D) none of the mentioned
9. **The basic figures of merit for MOS devices are**
A) Minimum Feature size
B) Low Power dissipation
C) Maximum operational frequency
D) All of the mentioned
10. **The scaling factor of Gate capacitance is:**
A) $1/\beta$
B) $1/\alpha$
C) β/α^2
D) α/β^2
11. **As the channel length is reduced in a MOS transistor, depletion region width must be**
A) Increased
B) decreased
C) must not vary
D) exponentially decreased

12. The junction parasitic capacitance is produced due to _____
 A) Source diffusion regions B) Gate diffusion regions
 C) Drain diffusion region D) All of the mentioned
13. Gate delay is given by
 A) R_{on}/C_g B) $R_{on} * C_g$
 C) C_g/R_{on} D) C_g^2/R_{on}
14. What is the relationship between channel resistance and sheet resistance where $Z=L/W$?
 A) $R = R_s$ B) $R = Z * R_s$
 C) $R = Z/R_s$ D) $R = R_s/Z$
15. In CMOS inverter characteristics, if n-transistor conducts and has large voltage between source and drain, then it is said to be in _____ region.
 A) Linear B) saturation
 C) non saturation D) cut-off
16. Increasing fan-out _____ the propagation delay in CMOS inverter
 A) increases B) decreases
 C) does not affect D) exponentially decreases
17. In CMOS logic circuit the n-MOS transistor acts as:
 A) Load B) Pull up network
 C) Pull down network D) Not used in CMOS circuits
18. Which color is used for n-diffusion representation in stick diagram?
 A) Red B) Blue
 C) Green D) Yellow
19. In CMOS domino logic _____ is possible.
 A) inverting structure B) non inverting structure
 C) inverting and non inverting structure D) very complex design
20. If the number of n selected input lines is equal to 2^m then it requires _____ select lines in multiplexer
 A) 2 B) M
 C) N D) 2^n

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT	CO	PO*
1. a)	State and explain Moore's law on the number of transistors on an IC.	4	L*1	1	1,2,5,12
b)	Explain body effect in MOS transistors.	5	L2	2	1,2,5,12
c)	Derive an expression for I_D in NMOS transistor.	7	L3	2	1,2,5,12
2. a)	Briefly explain the Flat Band Voltage in MOS transistors.	4	L2	2	1,2,5,12
b)	Compare MOS transistor with Bipolar transistor.	5	L2	1	1,2,5,12
c)	Explain the NMOS fabrication process.	7	L2	1	1,2,5,12
3. a)	Explain the drain current I_D vs drain to source voltage V_{ds} Characteristic.	4	L2	2	1,2,5,12
b)	Consider a MOS system with the following parameters: $N_A=3 \times 10^{16} \text{ cm}^{-3}$, $t_{ox}=200\text{\AA}$, $\epsilon_{ox}=3.9 \times 8.85 \times 10^{-14} \text{ F/cm}$, $\epsilon_{Si}=11.7 \times 8.85 \times 10^{-14} \text{ F/cm}$ and $q=1.6 \times 10^{-19} \text{ Coulomb}$ (i) Determine the substrate-bias (or body-effect) coefficient (ii) Determine surface potential with body at zero potential (iii) Determine the change in threshold voltage (V_T-V_{T0}) if $V_{SB}=2.5$	5	L3	2	1,2,5,12
c)	With the help of flowchart, discuss the VLSI design flow.	7	L3	1	1,2,5,12

Unit – II

4.	a)	List the short channel effects.	4	L1	3	1,2,5,12
	b)	Explain the latch up problem in CMOS inverter circuits.	5	L2	4	1,2,5,12
	c)	Derive an expression for pull-up to pull-down ratio ($z_{p.u}/z_{p.d.}$) for an NMOS inverter driven by another NMOS inverter.	7	L3	3	1,2,5,12
5.	a)	List the advantages and disadvantages of CMOS inverters over the other type of inverters.	4	L1	4	1,2,5,12
	b)	Explain constant voltage scaling in MOS transistor.	5	L2	3	1,2,5,12
	c)	Explain the method of estimating CMOS inverter delays i.e inverter rise and fall times.	7	L2	3	1,2,5,12
6.	a)	Explain the design rules for MOSFET layout.	8	L2	3	1,2,5,12
	b)	With help of a diagram, clearly explain the different operating regions of CMOS inverter.	8	L3	4	1,2,5,12

Unit – III

7.	a)	Design the nMOS based 2 input NOR gate and also provide stick diagram of the same. Also obtain expression for V_{OL} .	8	L4	5	1,2,5,12
	b)	Realise the expression of XOR gate using CMOS technology and draw its stick diagram.	8	L2	5	1,2,5,12
8.	a)	Design the CMOS and nMOS based 2 input NAND gate and also provide stick diagram of the same.	8	L4	5	1,2,5,12
	b)	Explain (i) Clocked CMOS logic (ii) CMOS domino logic.	8	L2	5	1,2,5,12

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
